

diffctx: Budgeted Typed-Graph Retrieval for Diff-Aware Code Context Selection

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May 2026

Abstract

Retrieving optimal context for Large Language Models to understand code changes is a critical challenge in automated software engineering. Current approaches rely on naive windowing or purely lexical retrieval, missing complex structural dependencies. We formulate diff-aware code context selection as **budgeted utility maximization** over multi-resolution fragments on a typed dependency graph. The deployed system maximizes a monotone submodular concept-coverage objective (concave saturation over per-concept relevance derived from the diff) with lazy-greedy budgeted selection and engineering heuristics; a modular cost-penalized (Boltzmann) ranking variant is available as an opt-in ablation, and we identify clean analyzable variants whose approximation guarantees are known. Two structural components ground the framework: (i) a partition matroid over multi-resolution fragments enforcing structural non-redundancy, and (ii) a pluggable relevance signal $\hat{w}(f, \Delta)$ instantiated via three families—bounded ego-network expansion (the deployed default), Personalized PageRank, and BM25 lexical retrieval. The framework is evaluated by file-level recall against annotated golden contexts at a fixed token budget across three multi-language software-engineering benchmarks (1500 instances total). At an 8000-token budget under bounded ego-network scoring, diffctx achieves pooled file recall 0.833 [95% CI 0.816, 0.850], with per-benchmark recall 0.913 on SWE-bench Verified, 0.847 on PolyBench-500, and 0.739 on ContextBench Verified. A paired-bootstrap comparison against a same-budget BM25-over-patch-identifiers baseline shows pooled diffctx recall 0.833 vs. BM25 0.566 (paired $\Delta \approx +0.267$, 95% CI [+0.245, +0.290], Wilcoxon $p < 10^{-85}$ on $n = 1500$ paired instances). An Aider repo-map baseline (fair-input mode) at the same $B=8000$ budget achieves pooled file recall 0.457 [0.434, 0.481] (per-benchmark: SWE-bench Verified 0.657, ContextBench Verified 0.432, PolyBench-500 0.284); the paired-bootstrap delta against diffctx-EGO is +0.376 [+0.348, +0.403], $p < 10^{-94}$ ($n=1500$). A three-mode scoring ablation across {EGO, PPR, BM25} over the budget grid {0, 8k, 16k, 32k, 64k, 128k, -1} is reported in Section 6.

1 Introduction

Large Language Models have demonstrated remarkable capabilities in code understanding tasks, yet their effectiveness depends critically on the context provided within limited token budgets. When reviewing code changes, developers and AI systems alike must determine which surrounding code fragments are necessary to understand the implications of a diff. Too little context leads to misunderstanding; too much context wastes tokens and introduces noise.

Current approaches to context selection suffer from fundamental limitations. Fixed-window methods (e.g., ± 10 lines around changes) ignore semantic structure and miss distant dependencies. Whole-file inclusion is wasteful and scales poorly. Lexical retrieval (BM25, embeddings) can miss structural relationships that are often important for code comprehension [13, 20]. None provide principled stopping criteria or approximation guarantees.

We address this gap by formalizing diff context selection as **budgeted submodular maximization** over Code Property Graphs. Our key insight is that “understanding a change” can be operationalized as covering explanatory concepts—definitions, call targets, type signatures, impacted tests—that propagate through structural dependencies with diminishing returns.

Task scope: post-hoc diff understanding. We address *post-hoc diff understanding*: a patch Δ is given as input, and the task is to retrieve a compact set of explanatory code fragments needed to review or reason about that patch within a token budget. This is distinct from *pre-resolution issue resolution* (the standard SWE-bench setup), where an agent must produce Δ from a problem statement without seeing the gold patch. The benchmarks in Section 5 are repurposed from issue-resolution datasets: we use their gold patches as known diffs against which file-level retrieval is scored, never as a target an agent must generate. The patch is on the input side; the recall metric measures whether the retrieved context covers the files the patch touches and the dependency neighborhood needed to interpret them.

Contributions.

1. A formal problem statement casting context selection as budgeted utility maximization over a partition matroid on multi-resolution fragments. We do not claim a new approximation algorithm; we use known optimization structure to make context selection explicit, analyzable, and extensible. An “algorithm $\times$ constraint $\times$ guarantee” table (Table 1) separates the deployed heuristic from analyzable variants; the deployed objective is the submodular concept-coverage form of Section 4.5
2. A typed-edge dependency-graph framework with per-category extraction and treatment (e.g., hub-suppression exemption), generalizing flat code-graph approaches
3. A pluggable relevance-scoring abstraction $\hat{w}(f, \Delta)$ where three algorithm families (bounded ego-network expansion, Personalized PageRank, BM25) instantiate the abstract signal under a single retrieval framework
4. An evaluation protocol using file-level recall against annotated golden contexts at a fixed token budget on multi-language software-engineering benchmarks, with paired-bootstrap statistical comparison against external baselines (BM25 over patch identifiers; Aider repo-map)
5. A reference implementation with tree-sitter parsing for 42 source and configuration formats and dedicated semantic edge builders for 36 of them (Appendix D). Of these, eight (Python, JavaScript, TypeScript, Java, Go, Rust, C, C++) appear in the empirical evaluation with $n \geq 10$ instances; the rest are supported but unvalidated at scale (Section D, Table 14)

2 Related Work

Program Dependence and Slicing. Program slicing [23] computes the subset of statements affecting a variable, forming the foundation for dependency-based code understanding. Program Dependence Graphs (PDGs) [6] and their interprocedural extensions [10] capture control and data dependencies, enabling precise impact analysis. Our work extends these ideas to context selection for LLMs, treating dependency graphs as the substrate for relevance propagation.

Developer Navigation and Information Foraging. Studies of how developers navigate code [13, 20] reveal that structural relationships (calls, definitions) dominate over lexical search. This informs our edge weighting scheme, prioritizing structural signals while retaining lexical features as fallback.

Submodular Optimization. Monotone submodular maximization admits efficient greedy approximations [18], with lazy evaluation [16] and stochastic variants [17] improving scalability. We apply this framework to context selection, modeling coverage of explanatory concepts as a submodular function.

Retrieval for Code Understanding. Recent work explores retrieval-augmented generation for code tasks, typically using embedding similarity or BM25 over code chunks. Our approach differs by grounding retrieval in explicit dependency structure rather than purely lexical or semantic similarity, and by providing a principled utility function for budget allocation.

Repository-Level Context Tools. Aider [7] proposes a related repository-mapping approach using Personalized PageRank over tree-sitter-extracted definition–reference graphs, with binary-search packing under a token budget. Our framework differs in three respects: (i) we formalize selection as budgeted utility maximization over a partition matroid with multi-resolution fragments, identify analyzable variants of the formulation whose approximation guarantees are known, and clearly separate the deployed heuristic from those analyzable variants, rather than presenting a heuristic packing without the formulation; (ii) we introduce a typed edge taxonomy with per-category treatment (e.g., hub-suppression exemption for semantic, structural, and test edges), and (iii) we treat scoring as a pluggable component, comparing PPR, ego-network expansion, and lexical baselines empirically. Sourcegraph Cody [8] formalizes code retrieval as a multi-stage RecSys pipeline with complementary retrievers (keyword, embedding, graph), but does not commit to a formal optimization framework. RepoMapper [5] replicates Aider’s PageRank approach as an MCP server.

Concurrent Work on Submodular Code Context. Concurrent work by Arafat [1] explores submodular packing for citation-grounded comprehension on conversational queries over Python repositories, combining BM25, dense embeddings, and graph expansion via import relations; their evaluation reports completeness improvements over plain greedy via submodular packing. Our work differs in three concrete respects: (i) diff-aware setting with explicit core edit set E_0 and budget reservation, versus their conversational query setting; (ii) typed edge taxonomy with ten categories and per-category treatment, versus their flat graph; (iii) explicit matroid–knapsack analysis with temperature reweighting, versus their direct submodular maximization over a single retrieved candidate set. The two works are complementary and together support the broader case for submodular formulations in code context selection.

Impact Analysis and Co-change. ATHENA [25] combines dependency graphs with transformer embeddings for impact analysis on bug-fix prediction; Ripple [24] extends this with LLM reasoning over evolutionary coupling. These works target predicting which files *will be*

changed given a seed, an adjacent task to context selection. DraCo [4] replaces random walks with static dataflow analysis for retrieval, demonstrating that explicit structural reasoning can outperform learned propagation in code-specific settings. We share their structural emphasis but differ in objective (context for understanding rather than prediction of next change) and formal framework.

3 Problem Formulation

3.1 Inputs and Definitions

Given repository states $V_{\text{before}}, V_{\text{after}}$ with diff $\Delta = \text{diff}(V_{\text{before}} \rightarrow V_{\text{after}})$, token budget B , and (during evaluation) an annotated golden set $G \subseteq F$ of fragments deemed relevant for understanding Δ , we seek a fragment set $C^* \subseteq F$ that maximizes weighted recall against G —approximated at inference by a hand-constructed relevance estimator $\hat{w}(f, \Delta)$ (Section 4.2)—subject to a partition matroid structure on F and a token budget $\text{cost}(C) \leq B$.

Fragment Universe F . The set of candidate fragments is defined as

$$F = \{(\text{source}, \text{start}, \text{end})\}, \quad (1)$$

contiguous, syntactically or semantically delimited regions of the sources, each carrying a stable identity. The notion is domain-agnostic: in source code, fragments are named declarations and their bodies (functions, types, modules, constants, rules); in prose, sections, paragraphs, or list items; in structured data (YAML, JSON, XML), keyed entries and array elements; in logs, individual events. Extraction is deferred to a domain-specific parser (tree-sitter for code, structural splitter for markup, regex-delimited for unstructured text), with indentation- or whitespace-based fallback when parsing fails. Fragments may also exist at multiple granularities for the same underlying unit—e.g., a full function body and a signature-only abridgement, or a full section and its heading—which introduces structural redundancy handled below.

Non-Redundancy via Partition Matroid $M = (F, \mathcal{I})$. Fragments in F may be mutually redundant: one may be nested inside another, multiple variants may exist at different granularities, or the same content may recur across documents. We define a pairwise redundancy relation $\sim$ on F (overlapping ranges, alternative resolutions of the same unit, duplicated content) and take its transitive closure to obtain an equivalence relation, which partitions F into classes $\mathcal{P} = \{P_u\}$. At most one representative per class may be selected:

$$\mathcal{I} = \{C \subseteq F : |C \cap P_u| \leq 1 \text{ for all } u\} \quad (2)$$

This is a structural hard constraint, preserved throughout selection up to a deliberate one-line boundary tolerance (back-to-back fragments may share a single delimiting line). The full-vs-abridged split for code is one instance of $\mathcal{P}$ (singleton-pair classes by construction); range-overlap-induced equivalence for general text is another (chains of overlapping fragments collapsed under transitive closure). In implementation, overlap detection over fragment ranges via a per-source sorted interval index suffices to enforce $\mathcal{I}$ without explicit matroid machinery.

Cost Model (modular proxy).

$$\text{cost}(C) = \sum_{f \in C} |f| \quad (3)$$

where $|f|$ denotes the size of fragment f in consumer-appropriate units (tokens for LLM consumers, bytes for storage-bound channels, lines for human readers), and any per-fragment metadata cost (source identifiers, position markers, delimiters) is folded into $|f|$ at extraction time. This is a *modular* approximation: the true assembled cost is bounded above by $\text{cost}(C)$, with overlap deduplication across nested fragments and shared per-source headers acting as a free reduction at materialization time. Modeling these second-order savings would render cost submodular and complicate optimization without changing feasibility. For LLM consumers, practical budgets range from 5k to 15k tokens, bounded by context window limits and the “lost in the middle” effect [14].

Knapsack Constraint. The cost model induces a knapsack constraint independent of the matroid:

$$\mathcal{K} = \{C \subseteq F : \text{cost}(C) \leq B\} \quad (4)$$

Feasible Region. The full feasible region is the intersection of matroid and knapsack:

$$\mathcal{F} = \mathcal{I} \cap \mathcal{K} = \{C \subseteq F : |C \cap P_u| \leq 1 \text{ for all } u, \text{ and } \text{cost}(C) \leq B\} \quad (5)$$

We separate $\mathcal{I}$ and $\mathcal{K}$ deliberately: the matroid is a hard structural constraint, while the knapsack is a numerical resource constraint. The literature provides several reference results for clean analyzable variants:

- **Modular function under a matroid:** greedy on the cost-blind weighted matroid problem returns the exact optimum (Edmonds; Rado–Edmonds matroid theorem), simply “one best representative per class”;
- **Monotone submodular under a cardinality constraint $|C| \leq k$:** greedy attains $(1 - 1/e)$ [18];
- **Monotone submodular under a general matroid (greedy):** $1/2$ [18] (Fisher–Nemhauser–Wolsey, the matroid theorem); $(1 - 1/e)$ via continuous greedy [2];
- **Monotone submodular under a knapsack:** $\frac{1}{2}(1 - 1/e)$ via density-greedy plus the arg max-singleton modification [12]; $(1 - 1/e)$ via partial enumeration [22];
- **Monotone submodular under matroid $\cap$ knapsack:** $(1 - 1/e - \varepsilon)$ via continuous greedy with contention resolution [3].

The deployed system is a heuristic and we do not invoke any of the above guarantees directly on it. Table 1 maps each variant the framework supports to its objective form, constraint, algorithm, and resulting claim.

Table 1: Algorithm $\times$ constraint $\times$ guarantee map. The deployed default is a heuristic instantiating the submodular concept-coverage objective of Section 4.5; analyzable variants of the framework admit the listed guarantees on their respective surrogate problems. The modular Boltzmann-ranked variant is opt-in.

Variant	Objective	Constraint	Algorithm	Claim
Deployed default (this paper)	monotone submodular concept coverage (Section 4.5) + adaptive stopping + rescue	budget $\cap$ partition matroid	lazy density-greedy with heuristics	empirical + additive stopping certificate (Prop. 1)
Modular cost-blind variant	modular	partition matroid	best representative per class	exact (Edmonds)
Modular + knapsack, arg max-singleton modification	modular	knapsack	modified density-greedy	$\frac{1}{2}$ (classical linear knapsack; $\frac{1}{2}(1 - 1/e)$ a fortiori from [12])
Clean submodular variant (Section 4.5, no stopping/bonuses)	monotone submodular	knapsack	modified density-greedy	$\frac{1}{2}(1 - 1/e)$ [12]; $(1 - 1/e)$ w/ enumeration [22]
Clean submodular variant, matroid only	monotone submodular	partition matroid	greedy; continuous greedy	$1/2$ greedy; $(1 - 1/e)$ continuous [18, 2]

Objective. The objective takes the same algebraic form in both modes but with different sources for the per-fragment weight $w(f, \Delta) \geq 0$:

$$\max_{C \subseteq F} \mathcal{U}(C) = \sum_{e \in C} w(e, \Delta) \quad \text{s.t.} \quad C \in \mathcal{F}. \quad (6)$$

Evaluation mode (golden set G available): $w(f) = \mathbb{1}[f \in G] \cdot \omega_f$, so the sum collapses to weighted recall against G :

$$\mathcal{U}_{\text{eval}}(C) = \sum_{e \in C \cap G} \omega_e. \quad (7)$$

Inference mode (no golden set): $w(f, \Delta) = \hat{w}(f, \Delta)$, a scoring function (Section 4.2) that estimates per-fragment relevance to Δ without access to G .

The modular form above is the analyzable simplification used in the guarantee discussion; in deployment the per-fragment relevance feeds the submodular concept-coverage aggregate of Section 4.5. Precision is controlled implicitly via the budget B , which bounds $|C|$. Asymmetric error cost—false negatives (missing a relevant fragment) degrade LLM understanding more severely than false positives—justifies the recall-centric formulation in both modes.

Modular vs. Submodular Instantiations. The framework supports both modular and submodular instantiations of $\mathcal{U}$ depending on how w is constructed. When w aggregates concept coverage with a concave saturation (Section 4.5)—*the deployed default in this work*— $\mathcal{U}$ is monotone submodular and the modified density-greedy of Section 4.6 carries the $\frac{1}{2}(1 - 1/e)$ guarantee under knapsack [12]; greedy under the partition matroid alone attains $1/2$ and continuous greedy attains $(1 - 1/e)$ [18, 2]. When w linearly weights fragments by relevance (the opt-in Boltzmann-ranked variant), $\mathcal{U}$ is modular: under a partition matroid alone the problem is exactly solvable by selecting the best representative per equivalence class (the trivial matroid optimum); under a knapsack the same modified density-greedy is $\frac{1}{2}$ -approximate on the modular-knapsack instance. We do not deploy continuous greedy. The optimization machinery is identical across modes; only the guarantee statement and the precise objective differ.

Pluggable Relevance Estimator. We treat $\hat{w}(f, \Delta)$ as a pluggable scoring component of the framework. In this work we instantiate it via three families: (i) *bounded ego-network expansion* with depth-decayed weights, the deployed default (Section 4.2.2); (ii) *Personalized PageRank* over a typed dependency graph (Section 4.2.1); (iii) *lexical similarity* via BM25 over fragment tokens (Section 4.2.3). The framework of this section is independent of which $\hat{w}$ is chosen; the analyzable variants of Table 1 apply uniformly to any non-negative scoring instantiation.

Cost-Penalized Surrogate (Lagrangian-Style). An optional cost-aware reweighting folds fragment cost into the per-fragment score:

$$\tilde{w}(f, \Delta) = w(f, \Delta) \cdot \exp(-\beta \cdot |f|), \quad (8)$$

yielding the surrogate objective $\tilde{\mathcal{U}}(C) = \sum_{e \in C} \tilde{w}(e, \Delta)$ over the partition matroid M . This is a Lagrangian-style relaxation: it solves a different penalized problem rather than the original hard-budget objective, and the post-hoc budget truncation that produces a feasible $C \in \mathcal{F}$ does not preserve the surrogate’s optimality. We use it as a budget-aware ranking signal during selection, not as an approximation algorithm for the original problem; we therefore do not claim that β -reweighting recovers an approximation guarantee for $\max \mathcal{U}$ subject to $\text{cost}(C) \leq B$. The parameter β is calibrated by binary search to hit a target expected cost on a validation set (Section 4.4); $\beta \rightarrow 0$ recovers pure relevance ranking, $\beta \rightarrow \infty$ collapses to cost minimization.

Core Edit Set E_0 . Constructed from diff Δ : for each hunk, the smallest covering fragment (else all overlapping fragments, else the nearest fragment) enters E_0 , together with the enclosing syntactic containers (functions/classes containing changes). Rather than assigning $w \rightarrow \infty$, the implementation places E_0 structurally before the greedy loop—equivalent to optimizing subject to $E_0 \subseteq C$, i.e. greedy on the objective contracted at E_0 (contraction preserves monotone submodularity)—subject to the budget reservation below. Symbol-level context makes relevance propagation robust to line renumbering.

Budget-Aware Core Selection. A secondary knapsack reservation

$$\text{cost}(C \cap E_0) \leq \beta_{\text{core}} \cdot B, \quad \beta_{\text{core}} \in [0.5, 0.7] \quad (9)$$

ensures the greedy expansion phase retains working budget for relevance-propagated context. If $\text{cost}(E_0) > \beta_{\text{core}} \cdot B$, lowest-priority core fragments are downshifted to a more compact representative within their equivalence class P_u (e.g., signature-only variant for code, heading-only for prose)—a legal swap inside the matroid—prioritized by relevance score. A rescue sweep then retries still-skipped cores (cheapest first, with signature fallback) against the full remaining budget, and any core that still does not fit is demoted to an ordinary greedy candidate. Without this constraint, large diffs consume the entire budget on core fragments and the optimization pipeline degenerates.

4 Methodology

4.1 Instantiation: Typed Dependency Graph for Source Code

We now instantiate the framework of Section 3 for source code. Relevance weights $w(f, \Delta)$ are derived from a typed weighted graph $G = (\mathcal{V}, \mathcal{E}, w, \tau)$ where nodes $\mathcal{V}$ correspond to fragments in F , edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ encode directed dependencies, $\tau : \mathcal{E} \rightarrow T$ assigns each edge a type from a fixed set T , and $w : \mathcal{E} \rightarrow \mathbb{R}_{\geq 0}$ assigns a per-type weight (Section 4.3). Relevance is propagated by Personalized PageRank seeded at the diff (Section 4.2.1).

Pipeline phases. The implementation realizes Section 3’s formalism in two preliminary phases before any scoring runs. *File discovery* (`candidate_files.rs`) walks the working tree under `.gitignore`-aware filters and assembles the candidate file set used as the source of $\mathcal{V}$; this resolves the implicit “which files do we even consider” question that the formalism leaves to the implementation. *Fragment extraction* (`parsers/`, `fragmentation.rs`) then tree-sitter-parses each file into the fragments of Table 2, producing the fragment universe F . These two phases are distinct: file discovery operates on paths (no AST work), fragment extraction operates on file contents (no I/O). A separate *symbol-level discovery* step (`discovery.rs`) seeds graph traversal from rare diff identifiers; it is part of the scoring pipeline (Section 4.2), not the universe-construction phases.

Fragment Types. Tree-sitter extraction yields 21 fragment kinds (Table 2), grouped by role: *full semantic units* (function bodies, class/struct/interface/enum/impl/type/module/variable/property/record definitions and declarations), *signature-only abridgements* of those (used as compact representatives in the partition matroid of Section 3), and *generic fallback* variants (Section, Chunk) for non-code text or files where parsing fails. Methods are encoded as **Function** fragments with the enclosing class providing context; only their signature variant (**MethodSignature**) is distinguished. Full units and their signature counterparts form the equivalence classes P_u of the matroid: at most one resolution per unit is selected.

Group	Variants
Full semantic units	Function, Class, Struct, Interface, Enum, Impl, Type, Module, Variable, Property, Record, Declaration, Definition
Signature-only	FunctionSignature, MethodSignature, ClassSignature, StructSignature, InterfaceSignature, EnumSignature
Generic fallback	Section, Chunk

Table 2: Fragment kinds extracted by the implementation. Full units and their signature counterparts form partition-matroid equivalence classes; generic fallback variants are used when tree-sitter parsing is unavailable or for non-code text.

Edge Categories. The set T comprises ten categories (Table 3): three core static-dependency categories (**Semantic**, **Structural**, **Similarity**); a **Sibling** category for file co-location (same-directory coupling); four extension categories that go beyond pure source-code dependencies (**Config**, **ConfigGeneric**, **Document**, **TestEdge**); the empirical **History** category mining co-change patterns from the git log [9]; and a **Generic** fallback for unclassified relations. **Config** subsumes domain-specific infrastructure cross-references (Helm charts, Kubernetes manifests, Dockerfile/Compose dependencies, CI/CD pipeline links, build-system configs); these share

the same category multiplier per the per-category parameterization (Section 4.3). **Semantic** subsumes call/import/type/symbol-reference edges, which are individually distinguished (and individually weighted) at extraction time but share a single category multiplier. **TestEdge** links tests to the code they exercise via static naming and import relations; the system does not execute tests, so runtime pass/fail signal is unavailable and weighting test edges by observed failures is future work. Three categories—**Semantic**, **Structural**, and **TestEdge**—are exempt from hub-suppression on the basis that their edges encode load-bearing structural and behavioral relationships that should not be dampened by node degree.

Category	Description
Semantic	Symbol references: function calls, identifier uses, type references, imports, Terraform module references.
Structural	AST containment (parent–child scope nesting).
Sibling	Files in the same directory (weak co-location coupling).
Similarity	Lexical/token overlap (TF-IDF on identifier tokens).
Config	Domain-specific infrastructure cross-references: Helm charts, Kubernetes manifests, Dockerfile/Compose dependencies, CI/CD pipelines, build-system configs (Bazel, CMake, etc.).
ConfigGeneric	Generic YAML/JSON cross-references not covered by domain-specific extractors.
Document	Cross-references in documentation (markdown links, reference-style citations).
History	Co-change patterns mined from git history.
TestEdge	Test-to-code relationships.
Generic	Fallback for unclassified edges.

Table 3: Edge categories in the typed dependency graph. Per-category multipliers w_τ default to 1.0 pending calibration (Section 4.3); the structural-over-lexical prior lives in fine-grained per-edge-type weights.

Symbol Resolution Caveat. **Semantic** edges (call/import/type/symbol references) use AST queries and intra-scope lexical name matching as a heuristic approximation to data-flow analysis. True def-use chains require control-flow graph construction and reaching-definitions analysis, infeasible for incomplete or unparsable code in git working trees. Static resolution coverage is codebase-dependent and degrades with metaprogramming and dynamic dispatch (more severely in dynamic languages, less so in statically typed code with explicit annotations); quantifying resolution recall against runtime call graphs is left to empirical validation.

4.2 Relevance Scoring: A Pluggable Component

The framework of Section 3 is agnostic to the choice of $\hat{w}(f, \Delta)$. We instantiate it via three families spanning the design space—bounded local expansion (deployed default), global graph propagation, and pure lexical retrieval. All three are first-class scoring modes in the implementation; fine-grained per-edge-type weights are held fixed at literature-derived priors across modes (Section 4.3). Empirical comparison is reported in Section 5.

4.2.1 Personalized PageRank

We compute relevance scores via random walk with restart from core edits E_0 . Let s denote the restart distribution: $s(v) = 1/|E_0|$ for $v \in E_0$ and $s(v) = 0$ otherwise; s is a probability

distribution by construction. The PPR fixed point is then

$$R_{\text{PPR}}(v) = (1 - \alpha) \cdot s(v) + \alpha \cdot \sum_{(u \rightarrow v) \in \mathcal{E}} R_{\text{PPR}}(u) \cdot \frac{w_{uv}}{\deg_{\text{out}}(u)} \quad (10)$$

where $\deg_{\text{out}}(u) = \sum_{u \rightarrow x} w_{ux}$ is the weighted out-degree, ensuring outgoing transition probabilities sum to 1.

The damping factor $\alpha \in [0.5, 0.65]$ controls locality. With continuation probability α , the number of *traversed edges* per walk before a restart follows a geometric distribution on $\{0, 1, 2, \dots\}$ with mean $\alpha/(1 - \alpha)$; equivalently, the number of *visited nodes* (including the seed) is $1/(1 - \alpha)$.¹ For $\alpha = 0.6$ this gives 1.5 traversed edges (or 2.5 visited nodes) before restart, sharper locality than web search ($\alpha = 0.85$, ≈ 5.7 edges), appropriate for the tighter dependency structure of code graphs. The implementation computes the fixed point with a forward-push (local-push) solver rather than global power iteration: per-node residuals are pushed until every residual falls below 10^{-4} (an L_∞ bound on the residual vector) [15], with a global push cap ($100 \cdot n$, bounded at 2×10^6) guarding worst-case cost; runs that hit the cap are flagged as truncated and their relevance scores are biased toward the seeds.

Locality interpretation: In sparse code graphs with moderate branching, relevance decays approximately with graph distance, but the relationship is not strictly exponential on weighted directed graphs. Hub nodes with high in-degree can accumulate disproportionate mass regardless of semantic relevance, motivating the hub-suppression mechanism of Section 4.1.

Forward–Backward Blending. For diff understanding, both *forward* (callee-direction) and *backward* (caller-direction) propagation carry signal: forward PPR exposes the dependencies a changed function relies on, backward PPR exposes the call sites and tests impacted by it. The implementation runs PPR independently on the original graph and its transpose, then linearly blends the two scores:

$$R_{\text{PPR}}(v) = \rho \cdot R_{\text{PPR}}^{\text{fwd}}(v) + (1 - \rho) \cdot R_{\text{PPR}}^{\text{bwd}}(v), \quad \rho \in [0, 1] \quad (11)$$

where ρ is the forward-blend weight (default $\rho = 0.4$, i.e. backward-leaning, reflecting that for diff comprehension the impact-direction signal is typically stronger than the dependency-direction signal). The two PPR runs share the same α and convergence tolerance; only the edge orientation differs.

4.2.2 Bounded Ego-Network Expansion

Ego-network expansion computes relevance via depth-bounded BFS from the core edit set E_0 , with weights decaying by hop distance:

$$R_{\text{ego}}(v) = \sum_{u \in E_0} \mathbb{1}[d_{\text{hop}}(u, v) \leq L] \cdot \gamma^{d_{\text{hop}}(u, v)} \cdot W_{\text{path}}(u, v) \quad (12)$$

where $L \in \{1, 2\}$ is the expansion radius (`EgoGraphScoring.max_depth` in the implementation), $\gamma \in (0, 1]$ is a per-hop decay factor (`EgoScoringConfig.per_hop_decay`), $d_{\text{hop}}(u, v)$ is the number of edges in the shortest (by hop count) path from u to v in the edge-orientation-agnostic graph (BFS over the union of forward and reverse edges), and

$$W_{\text{path}}(u, v) = \max_{\pi \in \Pi_{=d_{\text{hop}}(u, v)}(u, v)} \prod_{(a, b) \in \pi} w_{ab} \quad (13)$$

¹Both quantities are reported in the literature; we adopt the edge-based convention throughout for consistency with the transition-matrix formulation.

is the maximum, over paths of shortest hop length, of the product of edge weights along the path. We use product aggregation along paths to model multiplicative attenuation: relevance through a chain of weak edges decays compoundingly, mirroring PPR’s transition-probability semantics. Sum aggregation is an alternative; we report the product variant in evaluation.

In addition to the graph term, the implementation adds a small bounded lexical tie-break: non-core fragments sharing identifiers with the diff receive a bonus of up to $\varepsilon = 0.1$ (DIFFCTX_EGO_LEXICAL_EPS; overlap capped at 10 identifiers). EGO scoring is therefore predominantly structural but not purely graph-only, and the EGO-vs-BM25 comparison in Section 6 should be read accordingly; setting $\varepsilon = 0$ yields a graph-only ablation configuration.

Unlike PPR, ego-expansion has no global stationary distribution: relevance is strictly local. This differs from PPR in two computational respects: (i) cost is $O(|E_0| \cdot \bar{d}^L)$ where $\bar{d}$ is average degree, typically faster than PPR’s iterative convergence; (ii) hub nodes do not accumulate mass through long random walks, eliminating part of the motivation for hub-suppression on the scoring side.

Hypothesis on locality. Files relevant to a code change tend to lie within a small graph radius of the seed edit; relevance signal beyond depth 2 is dominated by noise from utility hubs and broad-capture configuration files. PPR with high restart probability $\alpha \in [0.5, 0.65]$ effectively approximates EGO behavior, suggesting that for diff-aware context selection PPR’s global formulation may be over-engineered relative to the structural reality. Section 5 tests this empirically.

4.2.3 Lexical Baseline (BM25)

BM25 scoring treats fragments as documents and the diff as a query, ranking purely on lexical overlap of identifier tokens:

$$R_{\text{BM25}}(f) = \sum_{t \in q(\Delta)} \text{IDF}(t) \cdot \frac{\text{tf}(t, f) \cdot (k_1 + 1)}{\text{tf}(t, f) + k_1(1 - b + b \cdot |f|/|f|)} \quad (14)$$

with standard parameters $k_1 = 2.5$, $b = 0.75$. The query $q(\Delta)$ is the set of identifiers extracted from Δ . BM25 is the natural baseline: if structural reasoning provides no benefit, lexical retrieval should match graph-based methods. Reporting BM25 alongside graph methods is a methodological honesty check—without it, “graph-based context selection” is an unfalsifiable claim.

4.2.4 Discussion: What Each Mode Tests

The three scoring instantiations answer different questions in ablation:

- **BM25** establishes a lexical baseline and tests whether structural reasoning is necessary at all.
- **EGO** isolates the contribution of bounded local graph structure—“does the graph help, even just at depth 2?” This is the deployed default.
- **PPR** tests whether global relevance propagation provides additional signal beyond local expansion.

A monotone ordering $\text{BM25} < \text{EGO} < \text{PPR}$ would validate the structural-relevance hypothesis with maximum strength; non-monotone results (e.g., $\text{EGO} \approx \text{PPR}$ or $\text{EGO} > \text{PPR}$) inform task-structure conclusions about the radius of relevance in code change. Section 5 reports the comparison; the empirical result is $\text{EGO} \geq \text{PPR} > \text{BM25}$ on every benchmark, motivating EGO as the deployed default.

4.3 Edge-Type Weight Priors and Operational Calibration

We separate *structural* edge-type weights from *operational* hyperparameters of the selection algorithm.

Edge-type weights as fixed priors. Edge weights are parameterized in two layers. Fine-grained per-edge-type weights (e.g. call 0.65–0.75, inheritance 0.70–0.80, import 0.50–0.75, test-link 0.30–0.60, documentation 0.30; `config/weights.rs`) are derived from code-navigation literature priors [13, 20] and held fixed across modes and benchmarks; their ordering—structural and semantic relations above lexical ones—reflects the empirical finding that structural relationships dominate over lexical search during developer code navigation. A second layer of per-category multipliers $w_\tau \in \mathbb{R}_{\geq 0}^{|T|}$ (one per category of Table 3) currently defaults to 1.0 for every category, pending the calibration below. Full data-driven calibration over the $|T|$ -dimensional weight simplex,

$$w_\tau^* = \arg \max_{w_\tau \geq 0} \frac{1}{N} \sum_{i=1}^N \sum_{f \in C_i^*(w_\tau)} \mathbf{1}[f \in G_i] \cdot \omega_f, \quad (15)$$

is left to future work: the labeled corpora required to disambiguate ten weights without overfitting exceed the size used in the present evaluation. The above formulation is recorded here as the natural calibration target once such corpora become available; richer parameterizations such as learned per-edge weights $w(e) = \text{MLP}(\text{features}(e))$ over edge-context features remain a further direction.

Operational hyperparameters that are calibrated. The two scalar operational parameters of the selection algorithm are calibrated empirically:

- τ : the marginal-utility stopping threshold of the lazy-greedy loop (Section 4.6),
- β_{core} (a.k.a. `core_budget_fraction`): the fraction of the budget reserved for fragments adjacent to the core edit set E_0 before peripheral expansion.

These two parameters govern algorithm behavior at query time and admit a small grid search on a held-out manifest; calibration details are reported in Section 6. In the present work each scoring mode uses the same $(\tau, \beta_{\text{core}})$ values selected on the deployed-default (EGO) calibration; per-mode re-tuning is recorded as a sensitivity question in the limitations.

Per-source out-edge cap. The typed dependency graph admits power-law density on large repositories (django, mui/material-ui, microsoft/vscode), with raw edge counts reaching 3–5 M before any pruning. We cap each source node’s outgoing edge list to its top- K neighbors by post-suppression weight, applied *after* hub-suppression so the IDF damping computes against the full in-degree distribution. The default $K=64$ is the deployed value of the implementation; whether K admits a meaningful tuning gradient or can be treated as a structural constant is empirically open and we report a stratified analysis with paired statistical test in the appendix once a sufficiently powered comparison ($n \geq 500$ with Wilcoxon signed-rank) is completed. The implementation exposes the parameter via the `DIFFCTX_MAX_EDGES_PER_NODE` environment variable.

4.4 Temperature Calibration

The Boltzmann reformulation reduces the matroid+knapsack problem to a family of matroid problems parameterized by β : $\beta \rightarrow 0$ recovers cost-blind relevance maximization, $\beta \rightarrow \infty$ collapses to cost minimization. We propose a meta-level instance-conditional predictor:

$$\beta = f_\theta(B, \phi(\Delta, G_{\text{graph}})) \quad (16)$$

where features ϕ include diff token cost, fragment count, candidate-set size, graph density, and core-edit cost share. The predictor f_θ would be trained to minimize $|\text{cost}(C^*(\beta)) - B|$ on a validation corpus, with auxiliary recall floor against golden sets.

In the present work we describe the mechanism and implement deterministic calibration; we do not learn f_θ . The implementation selects β per-instance via geometric bisection over $[\beta_{\text{lo}}, \beta_{\text{hi}}]$ with the one-sided acceptance criterion $\text{cost}(C^*(\beta)) \geq B - \varepsilon$ (the symmetric $|\text{cost} - B| < \varepsilon$ collapses to this form because the budget cap saturates the upper side), which requires no training data. Replacement of binary search with a learned regression model is left to follow-up evaluation. Temperature calibration is a Lagrangian relaxation that affects budget targeting accuracy; it does not by itself recover any approximation guarantee on the original budgeted problem, and we do not claim such a recovery. The matroid-only $(1 - 1/e)$ row in Table 1 refers to continuous greedy on the matroid-only relaxation, not to the Boltzmann reweighting deployed here.

4.5 Submodular Extension via Concept Coverage

The modular weighted-recall objective generalizes to a submodular concept-coverage formulation, and this formulation is the *deployed default*: information needs Z (definitions, call targets, type signatures, impacted tests, background) are extracted from the diff by semantic analysis (`needs_from_diff` in the implementation), with no human annotation required. Trading modularity buys the ability to model *diminishing returns*: once a concept is well-covered by one fragment, additional fragments documenting the same concept add less value. The modular form is recovered as the special case $Z = G$, $\phi = \text{id}$, $\text{rel}(f, z) = \mathbf{1}[f = z] \cdot \omega_z$, which collapses the coverage sum to weighted recall against golden fragments.

The extension is defined by:

$$U(C \mid \Delta) = \sum_{z \in Z} \phi \left(\max_{f \in C} \text{rel}(f, z) \right) \quad (17)$$

where $\text{rel}(f, z) = R(f)$ if fragment f contains/defines concept z , and $\phi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a nondecreasing function (e.g., $\phi(x) = \sqrt{x}$ or $\phi(x) = \min(x, 1)$). Concave ϕ enforces saturation; identity ϕ recovers the modular case.

Theorem 1. *Let $a_{f,z} \geq 0$ for all $f \in F, z \in Z$, and let $\phi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be nondecreasing with $\phi(0) = 0$, under the convention $\max_{f \in \emptyset} a_{f,z} = 0$ (so $U(\emptyset) = 0$). Then $U(C) = \sum_{z \in Z} \phi(\max_{f \in C} a_{f,z})$ is monotone submodular.*

Proof. Fix concept z and define $g_z(C) = \phi(\max_{f \in C} a_{f,z})$. For $C \subseteq T$, let $m_C = \max_{f \in C} a_{f,z}$ and $m_T = \max_{f \in T} a_{f,z}$. By set inclusion, $m_C \leq m_T$.

The marginal gain from adding fragment v to C is:

$$\Delta_C(v) = \phi(\max(m_C, a_{v,z})) - \phi(m_C)$$

Case 1: If $a_{v,z} \leq m_C$, then $\Delta_C(v) = 0$.

Case 2: If $a_{v,z} > m_C$, then $\Delta_C(v) = \phi(a_{v,z}) - \phi(m_C)$.

For $T \supseteq C$: If $a_{v,z} \leq m_T$, then $\Delta_T(v) = 0 \leq \Delta_C(v)$. If $a_{v,z} > m_T$, then since $m_T \geq m_C$ and ϕ is nondecreasing:

$$\Delta_T(v) = \phi(a_{v,z}) - \phi(m_T) \leq \phi(a_{v,z}) - \phi(m_C) = \Delta_C(v)$$

Thus g_z is submodular. Since $U = \sum_z g_z$ with each $g_z \geq 0$, U is submodular (nonnegative sums preserve submodularity).

Monotonicity: Adding fragments can only increase $\max_{f \in C} a_{f,z}$, and ϕ nondecreasing implies $U(C \cup \{v\}) \geq U(C)$. $\square$

The submodular objective is compatible with the partition matroid but breaks the Boltzmann reweighting machinery, which assumes modularity for the post-hoc truncation argument. In submodular mode, optimization uses direct knapsack with modified-greedy (Section 4.6, Option B), yielding $\frac{1}{2}(1 - 1/e)$ rather than $(1 - 1/e)$. This concept-coverage objective with $\phi(x) = \sqrt{x}$ is the deployed default; the modular Boltzmann-ranked variant is opt-in via an environment flag (`DIFFCTX_OBJECTIVE=boltzmann`). An objective-level ablation—submodular coverage vs. modular relevance at matched budgets—has not yet been run and is recorded as future work.

File-Importance Prior for Impact Needs. For impact-type needs we apply a per-fragment importance prior $I(f) \in (0, 1]$ to downweight peripheral code: $m'(f, n) = m(f, n) \cdot I(f)$ when $n.\text{type} = \text{impact}$. Three categories are distinguished by directory pattern: generated code ($I = 0.10$), peripheral directories such as `examples/`, `demo/`, `vendor/`, `fixtures/`, `docs/` ($I = 0.15$), and default code ($I = 1.0$). Values reflect domain priors—generated code is mechanically derived from a schema and carries no human-intent signal; peripheral directories are typically out of scope for production review—and are not calibrated to the benchmark; a $\pm 25\%$ / $\pm 50\%$ perturbation sensitivity check is recorded as an open ablation. Submodularity is preserved since $I(f) \in (0, 1]$ is a per-fragment static scalar applied before the max aggregation in $\phi(\max_f \text{rel}(f, z))$. Other need types (definition, signature, test, invariant, background) use unmodified $m(f, n)$ since they admit no analogous production-vs-peripheral distinction.

4.6 Greedy Selection with Adaptive Stopping

We employ a density-based greedy algorithm with lazy evaluation [16] for computational efficiency.

Two regimes. The greedy procedure operates in two regimes corresponding to how the budget is enforced.

Regime A (cost-penalized surrogate ranking). Score fragments by the cost-penalized $\tilde{w}(e) = w(e) \exp(-\beta|e|)$ and rank within each partition class; the budget is enforced post-hoc by truncation. This is a Lagrangian-style ranking heuristic, not an approximation algorithm for the original budget-constrained objective; we report it as one operating mode.

Regime B (explicit knapsack, current implementation). Density-greedy on the concept-coverage $\mathcal{U}$ of Section 4.5 with $\text{cost}(C) \leq B$, plus two engineering bonuses: a modular structural bonus ($0.10 \cdot r_{\text{norm}}$ for fragments matching at least one need), which preserves submodularity, and a coverage-saturation (“diversity”) bonus whose per-fragment increment is pointwise non-increasing in the selected set but path-dependent, falling outside the strict set-function form (Heuristics Annex); setting `DIFFCTX_RELATEDNESS_BONUS=0` disables the path-dependent term for clean analyzable runs. Without the adaptive-stopping heuristic and bonuses, the modified-greedy variant (Option B below) admits $\frac{1}{2}(1 - 1/e)$ on the monotone submodular-knapsack instance [12]. With adaptive stopping (deployed default), the variant is heuristic; Proposition 1 bounds the utility foregone by stopping.

Regime B is the default path. Regime A is opt-in via the `DIFFCTX_OBJECTIVE=boltzmann` environment flag, applying β -reweighting with binary-search calibration over the modular relevance scores, used as a budget-aware ranking signal rather than an approximation guarantee. Comparison between the two regimes is empirical (Section 5).

Approximation guarantees. For monotone submodular maximization, the classical greedy algorithm achieves a $(1 - 1/e) \approx 0.632$ approximation under **cardinality constraints** ($|C| \leq k$) [18]. Our problem uses a **knapsack constraint** ($\text{cost}(C) \leq B$ with heterogeneous fragment

costs), for which density-greedy does not admit the same guarantee. We therefore present two options:

Option A (Heuristic): Use density-greedy as a practical heuristic without worst-case bounds. We hypothesize that performance on real code graphs may exceed 0.5 of optimal due to favorable structure, but this requires empirical validation.

Option B (Modified Greedy with Guarantee): Return the better of (i) the density-greedy solution C_{greedy} and (ii) the best single fragment fitting the budget:

$$C_{\text{out}} = \arg \max \{U(C_{\text{greedy}}), \max_{f: |f| \leq B} U(\{f\})\}$$

This standard modification provides a constant-factor approximation guarantee under knapsack constraints [12]. **Note:** this guarantee applies only to the variant *without* adaptive stopping; the stopping heuristic is evaluated empirically.

Algorithm 1 Lazy Greedy Context Selection (Heuristic Version)

```

1: Pack  $E_0$  (relevance-descending) under  $\beta_{\text{core}} \cdot B$ , downshifting oversized cores to signature
   variants; rescue-sweep skipped cores against the full budget;  $C \leftarrow$  packed cores
2:  $Q \leftarrow$  priority queue of all remaining candidates by marginal density
3:  $d_{\text{peak}} \leftarrow 0$ 
4: while  $\text{cost}(C) < B$  and  $Q \neq \emptyset$  do
5:    $f^* \leftarrow$  pop highest-density candidate from  $Q$ 
6:   if  $f^*$  evaluation is stale then
7:     Recompute density, reinsert into  $Q$ 
8:   continue
9:   end if
10:  if  $\text{density}(f^*) > d_{\text{peak}}$  then
11:     $d_{\text{peak}} \leftarrow \text{density}(f^*)$ 
12:  else if  $d_{\text{peak}} > 0$  and  $\text{density}(f^*) < \tau \cdot d_{\text{peak}}$  then
13:    break ▷ Adaptive stopping (heuristic; Prop. 1)
14:  end if
15:  if  $\text{cost}(C) + |f^*| \leq B$  then
16:     $C \leftarrow C \cup \{f^*\}$ 
17:  end if
18: end while
19: return better of  $C$  and the best feasible singleton (Option B)

```

Adaptive stopping (engineering heuristic). The stopping threshold $\tau \in [0.05, 0.15]$ (DEFAULT_STOPPING_THRESHOLD in `config/limits.rs`, overridable via the `--tau` CLI flag; deployed default 0.12) terminates expansion when marginal utility-per-cost falls below a fraction of the running peak density d_{peak} . This modification improves token efficiency in practice but forfeits the constant-factor guarantee against the offline optimum, as greedy analysis assumes selection continues to the feasibility frontier. It does, however, admit an additive certificate:

Proposition 1 (Additive stopping certificate). *Run density-greedy on a monotone objective whose per-candidate marginal gain is non-increasing in the selected set—true for the monotone submodular $\mathcal{U}$ of Section 4.5, and preserved by the deployed bonuses (the structural bonus is modular; the diversity bonus’s increment is pointwise non-increasing in C). Let d_{peak} be the maximum marginal density observed during the greedy expansion phase and B_{rem} the unspent budget at the stopping point. If the loop stops when the best available candidate has density below $\tau \cdot d_{\text{peak}}$, then continuing the same greedy to the feasibility frontier would increase the objective by less than $\tau \cdot d_{\text{peak}} \cdot B_{\text{rem}}$.*

Proof. At the stopping point every candidate’s current density is at most that of the popped best, hence below $\tau \cdot d_{\text{peak}}$. Marginal gains are non-increasing in the selected set, so any fragment f the continued greedy would accept is accepted at density below $\tau \cdot d_{\text{peak}}$, contributing marginal utility below $\tau \cdot d_{\text{peak}} \cdot |f|$. The accepted fragments’ total cost is at most B_{rem} ; summing yields the bound. $\square$

The certificate is relative to the greedy continuation, not to the offline optimum; the constant-factor discussion of Table 1 concerns the latter and applies to the variant without stopping. The implementation emits the bound per run (`stopping_certificate` = $\tau \cdot d_{\text{peak}} \cdot B_{\text{rem}}$ in the analytics output) whenever the loop terminates by threshold. A with-vs-without-stopping ablation at matched budgets has not yet been run and is recorded as future work. Recommended values: $\tau \approx 0.08$ for focused tasks (bug fixes); $\tau \approx 0.12$ for exploratory tasks (refactoring).

Complexity. The algorithm runs in $O(|\mathcal{V}| \log |\mathcal{V}| + k \cdot T_{\text{eval}})$ where $k = |C|$ (typically 15–50 fragments) and $T_{\text{eval}} = O(|Z|)$. Lazy evaluation significantly reduces recomputations versus standard greedy on sparse code graphs.

5 Evaluation Methodology

Since “understanding” is latent and task-dependent, we operationalize it via a behavioral proxy with directly measurable outcomes: *file-level recall against an annotated golden context* on multi-language software-engineering benchmarks. Section 6 reports results under this protocol; additional proxies (dependency coverage, co-change recall, fault-localization accuracy) are described as natural extensions in Section 8.1.

5.1 Primary Metric: File Recall at Fixed Token Budget

For a diff Δ with golden file set G and selector output C (the selected fragments projected to their containing files) subject to $\text{cost}(C) \leq B$, define

$$\text{Recall}(C, G) = \frac{|C \cap G|}{|G|}, \quad \text{Precision}(C, G) = \frac{|C \cap G|}{|C|}. \quad (18)$$

At a fixed token budget B , recall is the load-bearing metric: precision is constrained by $|G|/k$ where k is the number of files that fit, and is reported only for completeness. The metric is independent of which scoring instantiation $\hat{w}$ is active and is computed identically across all three modes (EGO, PPR, BM25) and across baselines.

5.2 Test Sets and Benchmarks

Evaluation uses three frozen manifests aggregating instances from publicly released benchmarks: SWE-bench Verified [11], a curated 500-instance subset of SWE-PolyBench, and the fragment-annotated ContextBench Verified configuration. Stratification is performed per (`source_benchmark`, `language`); cross-benchmark contamination is filtered by (`repo`, `base_commit`) key. Manifests are committed to git before any calibration trial executes. HuggingFace dataset revisions are pinned by SHA so that subsequent reruns operate on the same data.

5.3 Calibration Protocol

Two operational parameters are calibrated on a separate calibration manifest disjoint from the test manifests: τ (stopping threshold) and β_{core} (core budget fraction), as defined in Section 4.3. The calibration objective is the minimum file-recall across the four calibration source benchmarks,

$$\text{score}(\tau, \beta_{\text{core}}) = \min_{b \in \mathcal{B}_{\text{cal}}} \text{recall}_b(\tau, \beta_{\text{core}}), \quad (19)$$

chosen to avoid bias toward the largest-population calibration benchmark. A grid search of 4×3 values over $(\tau, \beta_{\text{core}})$ is then validated on a held-out 235-instance validation manifest before fixing parameters and running the test set exactly once.

5.4 Baselines

Two external baselines are implemented at the same fixed token budget B on identical manifests:

- **BM25 over patch identifiers** (`benchmarks/baselines/bm25_baseline.py`): rank-bm25 over all repo files, query = identifiers extracted from the gold patch, file-pack to budget. This isolates pure lexical retrieval without graph structure and corresponds to the SWE-bench paper’s reference protocol [11] adapted to a tight budget.
- **Aider repo-map** [7] (`benchmarks/baselines/aider_baseline.py`): the closest publicly available approach using the same primitives (tree-sitter AST extraction + PageRank + token-budgeted selection). To avoid an unfair information leak, we run Aider in two modes, both at the same $B=8000$ token budget on the same manifests:

- *Fair-input Aider*: `mentioned_fnames` restricted to file paths that already appear in the input diff text (the same information `diffctx` receives from the patch). `mentioned_idents` extracted from the patch identically to how it is computed for the BM25 baseline.
- *Oracle-mentioned Aider*: `mentioned_fnames` populated from `instance.gold_files`. This is **not** a baseline but an approximate upper-bound stress test — it is what Aider would produce if a perfect oracle named the gold files in advance.

The fair-input mode is the headline comparison; the oracle mode bounds how much downstream value file-hint information could provide to a PPR-on-tree-sitter approach.

Sourcegraph Cody [8] is cited as related work but cannot be directly compared: its evaluation suite is internal and requires the Sourcegraph backend.

6 Empirical Evaluation

This section instantiates the protocol of Section 5 and reports the v2 evaluation run. The diffctx-EGO (deployed default) run, the BM25 file-level baseline, and the Aider repo-map fair-input baseline have completed on the full 1500-instance test set; the Aider oracle-mentioned stress test remains queued.

6.1 Setup

Test sets. Evaluation uses three frozen manifests, 500 instances each (1500 total), stratified per (`source_benchmark`, `language`):

- *SWE-bench Verified* [11] — 500 Python instances drawn from 12 unique repositories.
- *PolyBench-500* — a 500-instance multi-language curated subset of SWE-PolyBench, spanning 21 unique repositories across Java (125), JavaScript (125), TypeScript (95), and additional languages.
- *ContextBench Verified* — 500 instances spanning 58 unique repositories with fragment-level golden annotations across 8 languages (Python 266, TypeScript 66, JavaScript 60, Go 40, C 23, Rust 20, Java 11, C++ 10, plus 4 unclassified).

Cross-benchmark contamination is filtered by (`repo`, `base_commit`) key (1449 instances dropped from the calibration pool). HuggingFace dataset revisions are pinned by SHA in `benchmarks/dataset_revisions.json`.

Calibration. A 4×3 grid over $\tau \in \{0.04, 0.08, 0.12, 0.16\}$ and $\beta_{\text{core}} \in \{0.5, 0.7, 0.9\}$ was run on the full v1 calibration manifest (2,119 instances spanning SWE-bench Lite, PolyBench, Multi-SWE-bench, ContextBench). The selection objective is the minimum file-recall across the four calibration source benchmarks (Section 5); minimum rather than mean prevents domination by the largest-population benchmark. The full grid is reported in Table 4. The selected cell is $\tau=0.12$, $\beta_{\text{core}}=0.5$, with $B=8000$ tokens. Two provenance caveats apply: the calibration sweep used the previous (now-removed) hybrid combiner, and the per-cell instance logs for the full grid were not retained in the repository—the grid scores and winner are recorded in the calibration commit (`3ecb8dc5`). A re-calibration on the v2 EGO-default pipeline, which would also regenerate the grid artifact, is scheduled and treated as an open item in Section 6.7. Three observations: first, the surface is nearly flat—the top-3 cells lie within 0.001 of each other (Table 4), indicating that $(\tau, \beta_{\text{core}})$ are robust rather than tuned to a sharp optimum; second, the absolute score 0.1092 is the minimum mean recall across benchmarks on a difficult cross-language calibration mix—this is a baseline number for downstream multi-hop work, not a final test-set headline; third, the per-benchmark breakdown at the winner is `{swebench_lite: 0.133, polybench: 0.109, multi_swebench: 0.173, contextbench: 0.127}`, showing the bottleneck benchmark is PolyBench. The remaining operational parameters (Section 4.6 and the appendix) are held fixed at the values listed there.

Reproducibility. Test, validation, and calibration manifests were frozen and committed to git (commit `5852d3bb`, “*feat(benchmarks): freeze v1 splits*”, 2026-04-29) before any calibration trial executed. Calibration trials checkpoint to disk; final test runs are conducted exactly once on the frozen splits. Stratification seed is 42 with validation fraction 10%; details in `benchmarks/manifests/v1/SPLIT_REPORT.md`. Hardware: Apple M4 Pro, 48 GB unified memory, macOS arm64; Python 3.12.12 with dependencies pinned in `requirements-bench.lock` (`uv pip compile`); Rust toolchain pinned to 1.92.0 via `rust-toolchain.toml`; per-instance timeout 180s; checkpoint/resume supported.

Table 4: Calibration grid scores: minimum per-source-benchmark file recall on the v1 calibration manifest (2,119 instances, four benchmarks, $B=8000$). The selected cell is highlighted. Surface is flat (top-3 within 0.001); τ varies modestly across rows, β_{core} is essentially insensitive within the winning row. This grid was originally run under the v1 hybrid combiner; re-calibration on the deployed-default EGO pipeline is scheduled (Section 6.7).

$\tau \setminus \beta_{\text{core}}$	0.5	0.7	0.9
0.04	0.1081	0.1068	0.1053
0.08	0.1091	0.1081	0.1050
0.12	0.1092*	0.1077	0.1050
0.16	0.1036	0.1029	0.1011

6.2 Headline Results: EGO Mode (Deployed Default)

Table 5 reports per-benchmark file-level recall and precision over the full sweep (all three test sets at 500/500). Table 6 breaks the same data down by source language with non-parametric percentile bootstrap 95% confidence intervals ($B=10,000$ resamples, seed=42).

Table 5: Per-benchmark file-level metrics, scoring=EGO (deployed default), $B=8000$ tokens, with 95% percentile bootstrap CIs ($B=10,000$ resamples, seed=42). Reported as *intent-to-treat*: recall and precision are computed over all $n=500$ instances per test set, with `clone_fail` and execution-error rows contributing recall 0. The `ok%` column reports the fraction of instances where the pipeline produced a non-error selection. An *ok-only* sensitivity analysis (recall over the `ok%` subset) gives 0.810 on ContextBench Verified, 0.935 on PolyBench-500, and 0.913 on SWE-bench Verified: infrastructure failures (clone, patch-apply, out-of-memory) depress the ITT headline by 7–9 recall points on the two multi-language sets, so the ITT numbers are a lower bound that conflates retrieval quality with runner robustness. We keep ITT as the headline for honesty. Macro and micro averages coincide because each test set has $n=500$.

Test set	n	File recall	File precision	ok%
ContextBench Verified	500	0.739 [0.708, 0.769]	0.131 [0.118, 0.144]	91.2%
PolyBench-500	500	0.847 [0.818, 0.876]	0.144 [0.129, 0.160]	90.6%
SWE-bench Verified	500	0.913 [0.889, 0.936]	0.069 [0.060, 0.078]	100.0%
Pooled	1500	0.833 [0.816, 0.850]	0.115 [0.107, 0.122]	93.9%

6.3 Observations

Recall at a tight 8k-token budget. Pooled mean file recall is 0.833 [95% CI 0.816, 0.850] across all 1500 instances at $B=8000$. Per-benchmark recall is highest on SWE-bench Verified (0.913), then PolyBench-500 (0.847), and lowest on ContextBench Verified (0.739); the per-set spread (about 17 percentage points) is much larger than each set’s CI half-width and is therefore benchmark-driven, not noise. SWE-bench Verified file recall plateaus at 0.913 across the entire budget grid {8k, 16k, 32k, 64k, 128k, -1 } and is identical for both graph-based scoring modes (EGO and PPR both return 0.913); this saturation reflects an upper limit imposed by gold-set composition and seed coverage on that benchmark rather than a property of the scoring objective, and the budget curve is therefore most informative on ContextBench Verified and PolyBench-500.

Table 6: Per-language metrics with 95% percentile bootstrap CIs ($B=10,000$). Pooled across the three test sets, *intent-to-treat*: instances whose pipeline run failed (clone, apply, or out-of-memory kill) contribute recall 0. The ok-only column reports mean recall over the n_{ok} instances with a completed selection.

Language	n	n_{ok}	File recall (ITT)	File recall (ok-only)
Python	891	850	0.831 [0.808, 0.852]	0.871
TypeScript	191	170	0.852 [0.804, 0.897]	0.957
JavaScript	185	183	0.883 [0.847, 0.917]	0.893
Java	140	134	0.874 [0.822, 0.921]	0.913
Go	40	19	0.444 [0.299, 0.593]	0.935
C	23	23	0.754 [0.640, 0.857]	0.754
Rust	20	20	0.937 [0.887, 0.979]	0.937
C++	10	10	0.715 [0.583, 0.848]	0.715

Per-language pattern. On ok-only recall, TypeScript (0.957), Rust (0.937), Go (0.935), and Java (0.913) lead; JavaScript (0.893) and Python (0.871) follow closely; C (0.754) and C++ (0.715) trail with sample sizes too small for firm conclusions. Go’s low intent-to-treat number (0.444 [0.299,0.593]) is *not* a retrieval outlier: 21 of the 40 Go instances failed with out-of-memory kills during pipeline execution on large vendored-dependency repositories (`status=oom_kill` in the run checkpoints; edge materialization on this class reaches millions of candidate edges pre-cap) and contribute recall 0 to the ITT mean; over the 19 completed instances, Go recall is 0.935, second only to Rust among $n \geq 20$ languages. The open item is engineering—bounding peak memory of graph construction—not the symbol-reference heuristic. Across the completed instances, dynamic languages (Python, JavaScript) give up only 4–9 recall points to the statically-typed cluster, suggesting the symbol-reference heuristic, while imperfect for dynamic dispatch (Section 4.2), covers most cases in modern codebases.

Precision is uniformly low (≈ 0.12). Mean file precision is approximately $|G|/k$ where $|G|$ is the gold-file count per instance ($\sim 1\text{--}3$) and k is the number of files that fit the budget. Low file precision indicates that the selector retrieves many non-gold files under the fixed budget; downstream LLM context quality depends on both recall and noise, so file recall is a necessary but insufficient proxy. ContextBench Verified provides block-level (line-range) annotations beyond the file level; we plan to report block-level recall on that subset to characterize how concentrated the selected context is around the gold regions, and to bound the true noise floor implied by the file-precision number.

6.4 Baseline Comparisons

Two external file-level baselines run the same protocol (same manifests, same budget, same metric) so that the comparison is apples-to-apples. The BM25 baseline and the Aider fair-input baseline are reported below; the Aider oracle-mentioned mode (an upper-bound stress test, not a baseline) is still queued at the time of writing. Methodology follows the standard IR significance protocol [21]: per-instance paired bootstrap on the recall delta ($B=10,000$ resamples, percentile CI) and one-sided Wilcoxon signed-rank p -value (alternative: `diffctx > baseline`).

BM25 over patch identifiers. `rank-bm25` indexes all repository files; the query is the set of identifiers extracted from the gold patch (camelCase + snake_case decomposition, language-

keyword filter). Files are packed in BM25 rank order under strict greedy budget compliance. Implementation in `benchmarks/baselines/bm25_baseline.py`. This isolates pure lexical retrieval without any graph structure.

Aider repo-map. The reference open-source approach using the same primitives (tree-sitter AST + Personalized PageRank + token-budgeted packing). Invoked through an isolated `uv tool` subprocess with the upstream `aider-chat==0.86.2` package. We run Aider in two modes: *fair-input* (`mentioned_fnames` restricted to file paths visible in the input diff text, the same information `diffctx` receives) and *oracle-mentioned* (`mentioned_fnames` populated from `instance.gold_files`, an upper-bound stress test rather than a baseline). Implementation in `benchmarks/baselines/aider_baseline.py`.

Pairing. Both baselines run on the same 1500 instances under the same `--workers 1` configuration that `diffctx` uses; the comparison below is therefore on all 1500 paired instances rather than a both-OK subset (rows where either method failed are kept and contribute their reported recall, including any zero from a failed selection). This is the apples-to-apples intent-to-treat number; an additional both-OK sensitivity analysis is omitted because the failure rates are symmetric across methods on this run.

Table 7: `diffctx` (EGO, deployed default) vs. external baselines at $B=8000$ tokens, all 1500 paired instances. Δ is the per-instance paired delta in file recall (positive favors `diffctx`). 95% paired-bootstrap percentile CI on Δ ($B=10,000$ resamples, seed = 42); p -value from one-sided Wilcoxon signed-rank (H_1 : `diffctx` > baseline). Aider oracle-mentioned rows pending.

Test set	n	diffctx	baseline	Δ recall [95% CI]	Wilcoxon p
<i>vs. BM25 over patch identifiers</i>					
ContextBench Verified	500	0.739	0.485	+0.254 [+0.215, +0.293]	5.4e−28
PolyBench-500	500	0.847	0.672	+0.175 [+0.143, +0.207]	7.02e−22
SWE-bench Verified	500	0.913	0.540	+0.373 [+0.333, +0.413]	3.02e−41
Pooled	1500	0.833	0.566	+0.267 [+0.245, +0.290]	4.22e−86
<i>vs. Aider repo-map (fair-input mode)</i>					
ContextBench Verified	500	0.739	0.432	+0.307 [+0.262, +0.352]	6.5e−29
PolyBench-500	500	0.847	0.284	+0.564 [+0.516, +0.610]	1.4e−51
SWE-bench Verified	500	0.913	0.657	+0.257 [+0.209, +0.305]	1.7e−20
Pooled	1500	0.833	0.457	+0.376 [+0.348, +0.403]	9.3e−95
<i>vs. Aider repo-map (oracle-mentioned, upper-bound stress test — not a baseline) — queued</i>					
ContextBench Verified	500	0.739	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
PolyBench-500	500	0.847	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
SWE-bench Verified	500	0.913	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Pooled	1500	0.833	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

6.5 Scoring-Mode Ablation

Section 4.2 introduces three pluggable scoring modes (EGO, PPR, internal BM25). To attribute the source of recall, we run each of the three modes in isolation on the full 1500-instance test manifest with operational hyperparameters held at the EGO-default values ($\tau=0.12$, $\beta_{\text{core}}=0.5$, $B=8000$). Per-mode re-tuning is recorded as a sensitivity question — if the deployed-default hyperparameters disadvantage a single-mode run, that disadvantage is itself an argument against the single mode in deployment.

Section 4.2.2 predicts that PPR with high restart probability α approximates bounded-radius EGO. The ablation tests this empirically: a monotone ordering $\text{BM25} \prec \text{EGO} \prec \text{PPR}$ would validate the structural-relevance hypothesis with maximum strength; an $\text{EGO} \approx \text{PPR}$ or $\text{EGO} > \text{PPR}$ result informs conclusions about the radius of relevance for diff-aware context selection.

Table 8: Scoring-mode ablation at $B=8000$, deployed-default operational hyperparameters. File recall with 95% percentile bootstrap CIs over $n=500$ instances per test set, intent-to-treat. The first two columns are graph-based scoring instantiations of $\hat{w}(f, \Delta)$ inside the diffctx pipeline; EGO is the deployed default. The third column reproduces the external whole-file BM25 baseline of Table 7 as a reference point at the same budget; an internal-BM25 instantiation (BM25 fragments inside the framework’s selection machinery, distinct from whole-file packing) is implemented in `src/scoring.rs::BM25Scoring` but was not run as a separate sweep cell in this version and is left as future work.

Test set	n	EGO (default)	PPR	BM25 (ext. ref.)
ContextBench Verified	500	0.739 [0.708, 0.769]	0.708 [0.677, 0.738]	0.485 [0.451, 0.519]
PolyBench-500	500	0.847 [0.818, 0.876]	0.846 [0.817, 0.875]	0.672 [0.640, 0.704]
SWE-bench Verified	500	0.913 [0.889, 0.936]	0.913 [0.889, 0.936]	0.540 [0.497, 0.582]
Pooled	1500	0.833 [0.816, 0.850]	0.822 [0.805, 0.839]	0.566 [0.544, 0.587]

The ablation reveals two structural facts. First, on SWE-bench Verified both graph-based modes (EGO, PPR) report identical recall (0.913) at every budget from 8000 to -1 (Table 9); this saturation is consistent with a gold-set composition in which seed coverage by patch identifiers already accounts for the bulk of returnable mass, and the scoring objective only re-orders ties. Second, on the two benchmarks where scoring matters (ContextBench Verified, PolyBench-500), EGO matches or slightly exceeds PPR (PPR trails by ~ 3 points on ContextBench Verified, statistically tied on PolyBench-500); this empirically validates the prediction in Section 4.2.2 that bounded-radius EGO captures the bulk of relevance for diff-aware selection without the additional global mass that PPR distributes, and motivates EGO as the deployed default. We do not yet present an internal-BM25 ablation that would isolate the contribution of the graph-based scoring component from the framework’s selection machinery (fragmentation, partition matroid, lazy-greedy stopping) at constant inputs; that ablation is the natural next experiment and is treated as future work.

6.6 Budget Curve

To support claims about budget efficiency, the same protocol is repeated under EGO scoring (deployed default) at $B \in \{0, 8k, 16k, 32k, 64k, 128k, -1\}$ on the full 1500-instance test set. $B=0$ is a recall-floor sanity bound (must report exactly 0); $B=-1$ is the unlimited-budget ceiling. The curve is reported in Table 9.

For comparison, Table 10 reports the same budget sweep for the Aider repo-map fair-input baseline. The $16\times$ budget gap between the two protocols at the headline crossover is informative: diffctx-EGO at $B=8000$ achieves pooled recall 0.833, while Aider at $B=128,000$ reaches only 0.719; the lexical/structural primitives Aider’s repo-map builds on do not appear to compensate for the absence of typed-edge propagation even when granted an order-of-magnitude larger token budget.

The same sweep for the external BM25 baseline (Table 11) bounds the regime in which graph-based selection helps: whole-file lexical packing is far behind at tight budgets (0.566 vs. 0.833 pooled at $B=8000$) but overtakes diffctx-EGO once the budget is large enough to simply include most candidate files—pooled 0.846 vs. 0.844 at $B=32k$, 0.900 vs. 0.846 at 64k, 0.911 vs. 0.846 at 128k. The contribution of typed-graph selection is therefore specific to the tight-budget regime ($B \lesssim 16k$), which is exactly the regime motivating the problem (Section 3, cost model

Table 9: Budget curve: pooled file recall under EGO mode (deployed default) across the budget grid, on the full 1500-instance test set. *Mean used tokens* is the per-instance mean of selected-fragment token counts under `tiktoken o200k_base`; the gap to the nominal budget reflects the lazy-greedy stopping rule rather than the budget being the binding constraint.

Budget B	n	Mean recall [95% CI]	Mean used tokens
0	1500	0.000 [0.000, 0.000]	0
8,000	1500	0.833 [0.816, 0.850]	5,116
16,000	1500	0.839 [0.822, 0.855]	9,779
32,000	1500	0.844 [0.828, 0.860]	16,882
64,000	1500	0.846 [0.830, 0.863]	26,157
128,000	1500	0.846 [0.829, 0.863]	35,998
$-1 (\infty)$	1500	0.855 [0.838, 0.871]	169,083

Table 10: Budget curve for the Aider repo-map fair-input baseline at the same instance-level inputs. Per-benchmark recall and pooled recall with 95% percentile bootstrap CIs ($B=10,000$ resamples, seed=42), intent-to-treat over $n=500$ per test set ($n=1500$ pooled). $B=0$ is the recall-floor sanity bound; $B=-1$ is excluded as Aider does not accept an unlimited-budget directive.

Budget B	n	ContextBench Verified	PolyBench-500	SWE-bench Verified	Pooled
0	1500	0.000	0.000	0.000	0.000
8,000	1500	0.432 [0.396, 0.468]	0.284 [0.249, 0.320]	0.657 [0.617, 0.698]	0.457 [0.434, 0.481]
16,000	1500	0.509 [0.474, 0.544]	0.356 [0.319, 0.394]	0.766 [0.730, 0.801]	0.544 [0.521, 0.567]
32,000	1500	0.601 [0.566, 0.637]	0.401 [0.363, 0.440]	0.839 [0.806, 0.869]	0.614 [0.591, 0.636]
64,000	1500	0.662 [0.627, 0.696]	0.455 [0.416, 0.495]	0.897 [0.870, 0.921]	0.671 [0.649, 0.693]
128,000	1500	0.714 [0.680, 0.747]	0.527 [0.489, 0.566]	0.916 [0.892, 0.938]	0.719 [0.698, 0.739]

and “lost in the middle” [14]); at large budgets, recall saturates for any reasonable method and precision, not recall, becomes the binding concern.

6.7 Limitations of the Current Empirical Snapshot

The numbers in this section reflect the v2 sweep at commit 7458eced. The following items are honest open questions about the snapshot:

- **SWE-bench Verified file recall is saturated.** On SWE-bench Verified, both graph-based scoring modes (EGO and PPR) return 0.913 at every budget from 8000 to unlimited, while external BM25 produces a strictly monotone curve from 0.540 at $B=8000$ up to ~ 0.95 at $B=128,000$. The plateau is consistent with the observation that on this benchmark the gold file set is dominated by files that are direct seeds of the patch, so any retrieval method that returns all surviving seeds achieves the same recall regardless of subsequent expansion. SWE-bench Verified is therefore informative as a baseline-comparison metric (graph $\gg$ lexical at small budgets) but uninformative as a scoring-mode comparison.
- **PolyBench-500 fragment-level metrics are not reported.** PolyBench-500 ships a `modified_nodes` field encoding CST node paths (e.g., `file.java->program->class_declaration:Foo->method_declaration:bar`) rather than line ranges, and the current adapter does not yet resolve these symbolic paths to `(start_line, end_line)` tuples via a tree-sitter pass on the post-patch worktree.

Table 11: Budget curve for the external BM25-over-patch-identifiers baseline, same protocol (intent-to-treat, $n=500$ per test set). BM25 does not support $B=-1$ (it packs whole files in rank order to the budget).

Budget B	n	ContextBench Verified	PolyBench-500	SWE-bench Verified	Pooled
0	1500	0.000	0.000	0.000	0.000
8,000	1500	0.485	0.672	0.540	0.566
16,000	1500	0.625	0.781	0.756	0.721
32,000	1500	0.737	0.871	0.931	0.846
64,000	1500	0.806	0.925	0.970	0.900
128,000	1500	0.849	0.936	0.947	0.911

PolyBench-500 therefore contributes to file-level recall only; fragment-level and line-level metrics on this benchmark are deferred.

- Aider PolyBench-500 infrastructure failure rate.** Aider fair-input on PolyBench-500 has a 24.4% non-ok status rate at $B=8000$: 85 instances (17.0%) hit the 600s helper-process timeout, and 37 instances (7.4%) fail with `QueryError: Invalid node type at row 2, column 2: module` — a tree-sitter grammar version mismatch in the bundled `aider-chat==0.86.2` package against the installed parsers. The intent-to-treat number (0.284 pooled at $B=8000$) folds these failures in as recall 0; an *ok*-only sensitivity analysis on the same cells gives 0.375 on PolyBench-500, still well below external BM25 (0.673 on the same instances). For comparison, Aider’s *ok%* is 94.0% on SWE-bench Verified (mostly Python, the language Aider’s repo-map heuristics are most mature on) and 84.8% on ContextBench Verified. The Aider oracle-mentioned variant (`mentioned_fnames` populated from `instance.gold_files`, an upper-bound stress test rather than a baseline) is queued; whether Aider-oracle approaches diffctx-EGO recall is the open question.
- $(\tau, \beta_{\text{core}})$ **post-fix re-calibration.** The hyperparameter calibration in Section 4.6 pre-dates the seed/selection fixes that produced the v2 numbers (in particular the rescue pass for cores skipped from $\beta_{\text{core}} \cdot B$ and the inclusion of pure-rename new paths in the candidate set). A focused re-calibration sweep over $\tau \in \{0.04, 0.08, 0.12, 0.16, 0.20\}$ on the v2 pipeline is scheduled; we expect the surface to remain flat as in v1 (Table 4), but report the v2 calibration as an open item rather than relying on the v1 result.
- Go out-of-memory class.** 21 of 40 Go instances are killed out-of-memory during graph construction on large vendored-dependency repositories and score recall 0 under intent-to-treat (Section observations above). Bounding peak memory of edge materialization (currently up to millions of candidate edges before the per-source cap) is the open engineering item; until fixed, Go ITT numbers measure infrastructure, not retrieval.
- Promised ablations not yet run.** Two ablations referenced by the methodology are open: objective-level (submodular concept coverage vs. modular relevance at matched budgets, Section 4.5) and stopping (with vs. without adaptive stopping at matched budgets, Section 4.6). The clean-run flags (`DIFFCTX_OBJECTIVE`, `DIFFCTX_RELATEDNESS_BONUS`, `DIFFCTX_EGO_LEXICAL_EPS`) exist so both can be run without code changes.
- Per-future timeout.** The runner’s per-future timeout does not enforce in single-worker mode because `concurrent.futures.as_completed` applies a bulk deadline rather than per-call. About 2% of instances exceed the configured 180s; this increases wall-clock but does not corrupt results.

7 Discussion and Limitations

Language Coverage. The framework requires language-specific tooling for AST parsing and name resolution. The current implementation supports Python, TypeScript, JavaScript, Go, Rust, Java, C, C++, Ruby, Kotlin, and Scala via tree-sitter parsing, with Python and JavaScript receiving dedicated semantic analysis. Dynamic languages present challenges due to incomplete static analysis of dynamic dispatch.

Symbol Reference vs. Data-Flow. Our “symbol reference” edges use lexical name matching within function scope as a heuristic. This is **not** true data-flow analysis, which requires CFG construction and reaching definitions computation—infeasible for incomplete code in git working trees. We explicitly acknowledge this limitation.

Forward-Backward PPR Sensitivity. The implementation runs PPR on both the original graph and its transpose and linearly blends the two scores (Equation 11); for change-understanding the default is backward-leaning ($\rho=0.4$). The blend weight ρ has not been re-calibrated per language or per benchmark; sensitivity to ρ is a natural extension and is recorded as future work.

Hub Node Dominance. Utility classes with high in-degree (Logger, Config, Utils) can accumulate excessive PPR mass. Our implementation applies IDF-style damping $w'_{uv} = w_{uv} / \log(1 + \text{in_degree}(v))$ for nodes exceeding the 95th percentile in-degree, and a second, out-fan mechanism damping semantic edges from source nodes that reference three or more distinct files ($w/\sqrt{\text{fan}}$; Heuristics Annex). While this improves PPR distribution in preliminary testing, systematic validation across diverse codebases is required to confirm its effectiveness.

Configuration Files. Non-code files (YAML, JSON, TOML) lack structural dependencies. For these, lexical signals must dominate, with higher BM25 weights (0.4–0.5).

Parameter Sensitivity. While we provide operational ranges for parameters (α , τ , edge weights), optimal values are repository-dependent. We recommend calibration on 20–50 historical commits via grid search.

8 Conclusion

We have presented `diffctx`, a framework for diff-aware code context selection that formulates the problem as budgeted utility maximization over multi-resolution fragments on a typed dependency graph. The deployed system maximizes a monotone submodular concept-coverage objective with lazy-greedy budgeted selection and engineering heuristics, including an adaptive stopping rule that carries an additive loss certificate (Proposition 1). A pluggable scoring abstraction (EGO, PPR, BM25) allows the same selection algorithm to be instantiated with different relevance signals without changing the optimization machinery; bounded ego-network expansion (EGO) is the deployed default, motivated empirically by the ablation in Section 6. We do not claim a new approximation algorithm; we use known optimization structure to make context selection explicit, analyzable, and extensible.

Key contributions: (1) an explicit “algorithm $\times$ constraint $\times$ guarantee” map (Table 1) that separates the deployed heuristic from analyzable variants of the framework, with the submodular concept-coverage objective (Section 4.5) as the deployed default and the modular Boltzmann variant as opt-in; (2) a typed-edge dependency graph with per-category treatment (hub-suppression exemption for semantic, structural, and test edges); (3) empirical results (Section 6) showing pooled file recall 0.833 [95% CI 0.816, 0.850] at $B=8000$ on 1500 multi-language instances under the deployed-default EGO scoring mode (per-benchmark recall: SWE-bench Verified 0.913, PolyBench-500 0.847, ContextBench Verified 0.739), and paired-bootstrap comparisons against two same-budget baselines: against external BM25-over-patch-identifiers, pooled $\Delta = +0.267$ [$+0.245, +0.290$], one-sided Wilcoxon $p \approx 4 \times 10^{-86}$; against Aider repo-map (fair-input mode), pooled $\Delta = +0.376$ [$+0.348, +0.403$], $p < 10^{-94}$ ($n=1500$ in both). The Aider budget curve (Table 10) shows that even at $B=128,000$ pooled Aider recall reaches only 0.719, below the `diffctx`-EGO recall of 0.833 at the $16\times$ smaller $B=8000$ budget. We further report a three-mode scoring ablation (EGO, PPR, external BM25 reference at $B=8000$) showing $\text{EGO} \geq \text{PPR} > \text{BM25}$ with EGO and PPR both ~ 25 points above the lexical baseline pooled; and a budget curve $B \in \{0, 8k, 16k, 32k, 64k, 128k, -1\}$ showing recall reaches 0.833 at $B=8000$ and saturates by $B=128k$ (asymptote 0.855 at $B=-1$). Conversely, whole-file BM25 packing overtakes EGO beyond $\sim 32k$ tokens (Table 11), scoping the contribution to the tight-budget regime in which context selection is actually hard.

8.1 Future Work

Additional behavioral proxies. The current evaluation uses file-level recall against an annotated golden set. Three complementary proxies are natural extensions:

- **Dependency coverage recall.** For a diff Δ , define $\text{Coverage}(C, \Delta) = |\{d \in \text{deps}(\Delta) : \exists f \in C, d \text{ resolves in } f\}| / |\text{deps}(\Delta)|$, where $\text{deps}(\Delta)$ is the set of definitions of modified symbols, call targets, callers, types in signatures, and covering tests. This complements file recall by measuring whether the selected context resolves the symbols that appear in the diff [19].
- **Co-change recall.** For commits where files A and B change together, input only A ’s diff and measure whether B appears in C . Co-change has a 30–40% incidental-coupling false-positive rate [9]; the proxy is meaningful only after filtering to semantic modifications (≥ 5 substantive lines).
- **Fault-localization accuracy.** Given (Δ, C) , an LLM-based or human evaluator ranks suspicious lines for root-cause identification, scored by Mean Reciprocal Rank and Precision@ k [19].

Edge-weight calibration on labeled corpora. Section 4.3 fixes w_τ at literature-derived priors. A larger labeled corpus (e.g., a sweep across all six benchmarks combined) would enable the full $|T|$ -dimensional calibration of w_τ , with the formulation already laid out there. Richer parameterizations (per-edge MLP over edge-context features) become tractable once labeled data scales further.

Blend sensitivity, memory, and dynamic dispatch. Forward–backward PPR blending is already deployed (Equation 11, $\rho = 0.4$); per-language and per-benchmark sensitivity to ρ has not been studied and is a direct extension. The most acute engineering item is the Go out-of-memory class (Section 6.7): bounding peak memory of edge materialization would recover the 21 currently-failing Go instances whose completed peers score 0.935. Per-language semantic resolvers (e.g. for interface-method-set dispatch and metaprogramming-heavy code) remain valuable for the residual gap between dynamic and statically-typed languages on completed instances.

Acknowledgments

The reference implementation lives at <https://github.com/nikolay-e/diffctx>; the treemap-per product layer (<https://github.com/nikolay-e/treemapper>) builds on it.

A Parameter Guidelines

Parameter	Range	Selection Guidance
α (damping)	0.50–0.65	Start at 0.60; decrease for over-selection of distant code
τ (threshold)	0.05–0.15	0.08 for bug fixes; 0.12 for features (heuristic)
Budget B	5k–20k tokens	10k standard; scale with diff complexity
Call weight	0.7–0.9	Tune via cross-validation on historical commits
Lexical weight	0.1–0.2	Reduce if high false positive rate
Overhead	15–20 tokens	Measure actual serialization cost per fragment

Table 12: Operational parameter ranges with selection heuristics.

Implementation parameters not modelled in Section 3–4.6. The reference implementation exposes additional parameters that govern second-order behaviors not formalized in the body: a relevance-augmentation weight `UtilityConfig.eta` (0.20) blending raw PPR scores into match strength; a structural-bonus weight `UtilityConfig.structural_bonus_weight` (0.10) adding a small bonus when a fragment has any matched need; an r_{cap} calibration constant `UtilityConfig.r_cap_sigma` (2.0, in standard deviations) controlling the soft cap on per-fragment relevance contribution; a hunk-proximity decay system (`FilteringConfig.proximity_half_decay` = 50 lines, `FilteringConfig.definition_proximity_half_decay` = 5 lines, `UtilityConfig.proximity_decay` = 0.30) that boosts fragments physically near diff hunks; BM25 hyperparameters `Bm25Config.idf_smoothing` (0.5) and `Bm25Config.min_query_token_length` (3), with the average document length computed from the corpus; the Boltzmann calibration loop’s bisection bounds `BoltzmannConfig.beta_lo` (10^{-6}), `BoltzmannConfig.beta_hi` (1.0), iteration cap `BoltzmannConfig.bisect_iters` (24), and convergence tolerance `BoltzmannConfig.calibration_tolerance` (5% of budget); and the rescue-phase parameters `RescueConfig.budget_fraction` (0.05) and `RescueConfig.min_score_percentile` (0.80). These parameters are calibrated empirically on representative corpora and held fixed across modes; their role is operational rather than theoretical, so we do not model them in the body but document them here for reproducibility.

B Algorithmic Complexity

Component	Time	Space	Practical Scale
AST parsing	$O(n)$	$O(n)$	Tested to ~ 500 k LOC
Call graph	$O(n \log n)$	$O(n + m)$	Tested to ~ 200 k LOC
PPR (sparse)	$O(k \cdot m)$	$O(n + m)$	Tested to ~ 50 k nodes
Lazy greedy	$O(\mathcal{V} \log \mathcal{V})$	$O(n)$	Tested to ~ 5 k fragments

Table 13: Complexity analysis where n = fragments, m = edges, k = PPR iterations. Practical scale limits are estimates requiring validation; performance depends on repository structure and hardware.

C Heuristics Annex

The following tactics are useful in practice but lack strong theoretical or empirical grounding. They should be evaluated via ablation before deployment.

Relatedness (Diversity) Bonus. Fragments with relevance above a small threshold (≥ 0.03 , `NeedsConfig.min_rel_for_bonus`) receive an additional marginal gain $\text{rel} \times 0.25 \times \text{unsatisfied}(C)$, where $\text{unsatisfied}(C) = \max(0, 1 - \sum_z \text{cov}_z(C)/|Z|)$ is the fraction of concept mass not yet covered by the selected set. This keeps structurally related fragments in play early while fading the bonus as coverage saturates. Separately, a modular structural bonus of $0.10 \cdot r_{\text{norm}}$ applies to fragments matching at least one need (`UtilityConfig.structural_bonus_weight`). **Warning:** the diversity term is path-dependent (its increment depends on coverage at selection time) and falls outside the strict set-function form, so worst-case approximation guarantees do not apply to the deployed configuration; its increment is pointwise non-increasing in C , which is the property Proposition 1 needs. Setting `DIFFCTX_RELATEDNESS_BONUS=0` disables it. An ablation with vs. without the bonus is still required.

Hub Suppression. Apply IDF-style penalty to edges into high in-degree nodes: $w'_{uv} = w_{uv} / \log(1 + \text{in_degree}(v))$ when $\text{in_degree}(v)$ exceeds the 95th percentile; edges in the exempt categories (`Semantic`, `Structural`, `TestEdge`) are not damped by this in-degree mechanism. A second, out-fan mechanism damps `Semantic` edges from source nodes referencing ≥ 3 distinct files: $w' = w / \sqrt{\text{fan}}$, targeting broad-capture import hubs. Both are implemented globally during graph construction, before the per-source top- K cap.

Backward Edge Weighting. Weight caller $\rightarrow$ callee edges at $0.7\text{--}0.8\times$ of callee $\rightarrow$ caller edges to prioritize impact analysis over dependency tracing.

Test File Handling. Test-to-code edges are static: direct links (test imports/references the unit) carry weight 0.60, naming-convention links 0.50, and reverse (code $\rightarrow$ test) links 0.30 (`config/weights.rs`). The system does not execute tests; upweighting tests that fail after Δ (e.g. to 0.9) requires runtime signal and is future work.

Signature Variants. Selection is fragment-granular, so partial function bodies never appear; instead, every full semantic unit gets a signature-only companion fragment (bracket-balanced signature extent, falling back to the first two lines; `signatures.rs`) that serves as its compact matroid representative when the full body does not fit.

Post-Passes. After the main greedy loop terminates (either by exhausting the budget or by adaptive stopping, Section 4.6), three ordered post-passes run (`postpass.rs`, `pipeline.rs`): (i) a *coherence pass* (`coherence_post_pass`) scans the selected set for dangling references—selected fragments whose semantic-neighbor definitions are absent from the context—and pulls in the missing definitions against the full remaining budget; (ii) a *rescue pass* (`rescue_nontrivial_context`) spends a reserved fraction (5% of B , `RescueConfig.budget_fraction`) on high-relevance fragments from files not yet represented, gated by a percentile threshold on relevance (`RescueConfig.min_score_percentile = 0.80`) so it does not flood the context with low-confidence rescues; (iii) a *changed-files pass* (`ensure_changed_files_represented`) ensures, budget permitting, that every file touched by Δ appears at least as a compact representative. All three only add fragments within budget and matroid feasibility, so for a monotone objective they never decrease utility below the greedy value; they do relax the strict density-greedy ordering.

D Language Support

The reference implementation includes tree-sitter parsing for 42 source and configuration formats and dedicated semantic edge builders for 36 of them. Coverage is uneven: a handful of languages have been used extensively in the evaluation; most others are present at the parser/edge-builder level but unvalidated against a benchmark of meaningful size. Table 14 groups languages by tier of empirical exercise so the contribution and the open work are not conflated.

Tier	Languages
Tier 1 — empirically validated ($n \geq 10$ in evaluation)	Python ($n=891$), TypeScript ($n=191$), Java ($n=140$), Go ($n=40$), C ($n=23$), Rust ($n=10$)
Tier 2 — semantic edge builders, no large-scale empirical validation	C#, Kotlin, Ruby, PHP, Swift, Scala (generic), Haskell, Elixir, Erlang, OCaml, Clojure (generic); Lua, R, Perl, Bash/Zsh/Fish/Ksh (generic); Objective-C (scripting); HTML, CSS/SCSS/LESS; HCL/Terraform, CMake, Make, Nix (markup); Bazel, dbt, OpenAPI, Prisma, Protobuf, SQL (main)
Tier 3 — parser only, structural fragmentation without semantic edges	YAML, JSON, TOML
Generic fallback	Markdown, plain text, dotfiles, unparsed source or statement fragmentation)

Table 14: Tiered coverage of the reference implementation. Tier 1 has both edge builders and per-language recall numbers in Table 6. Tier 2 implements the same edge builders but is not exercised against a benchmark of meaningful size in this paper; its inclusion is a code-level claim, not an empirical one. Tier 3 supplies parsing only — sufficient to fragment files for selection but not to draw structural edges. Treat the count of supported formats as an implementation upper bound; only Tier 1 carries empirical evidence.

E Symbol-to-Code Map

This appendix records the correspondence between symbols in the body of the paper and their identifiers in the reference Rust implementation. The map is the durable contract between the formalism and the code: a symbol that appears in the paper must resolve to a named field, function, or module in the codebase, and any drift surfaces here. Paths are relative to `diffctx/src/`.

Paper symbol	Meaning	Code identifier	Source
α	PPR damping factor	PPRConfig.alpha	config/limits.rs
ρ	PPR forward-blend weight (Eq. 11)	PPRConfig.forward_blend	config/limits.rs
β	Boltzmann inverse temperature	BoltzmannConfig.beta_lo/beta_hi (bisection bounds)	config/selection.rs
β_{core}	Core edit budget fraction	SelectionConfig.core_budget_fraction	config/selection.rs
γ	Ego-network per-hop decay (Section 4.2.2)	EgoScoringConfig.per_hop_decay	config/scoring.rs
τ	Adaptive stopping threshold	DEFAULT_STOPPING_THRESHOLD (CLI --tau)	config/limits.rs
L	Ego-network expansion radius	EgoGraphScoring.max_depth	scoring.rs
k_1, b	BM25 saturation, length norm	Bm25Config.k1, Bm25Config.b	config/bm25.rs
w_τ	Per-edge-category weight	CategoryWeights.<variant>	config/category_weights.rs
$I(f)$	File-importance prior	utility::importance::compute_file_importance	utility/importance.rs
ω_e	Per-fragment importance scaling	compute_seed_weights	core.rs
E_0	Core edit set	identify_core_fragments	core.rs
F	Fragment universe	output of fragment-extraction phase	fragmentation.rs, parsers/
<i>Pipeline phases:</i>			
File discovery	Candidate file assembly	candidate_files::collect_candidate_files	candidate_files.rs
Fragment extraction	Tree-sitter fragmentation	fragmentation::process_files_for_fragmentation	fragmentation.rs
Symbol-level discovery	Seed expansion via rare idents	discovery::*Discovery	discovery.rs
Scoring	Pluggable $\hat{w}(f, \Delta)$	PPRScoring, EgoGraphScoring, BM25Scoring	scoring.rs
Filtering	Relevance/proximity filters	filtering::*	filtering.rs
Selection	Lazy greedy / Boltzmann	select::lazy_greedy_select	select.rs
Coherence post-pass	Rescue dangling references	postpass::coherence_post_pass	postpass.rs

Table 15: Symbol-to-code map. Paper symbols on the left correspond to the named code identifiers on the right, located in the listed source file under `diffctx/src/`. Implementation-only parameters are documented inline in Appendix A and are not duplicated here.

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